

Passive Thermal Design of a 6U Small Satellite Under a High-Beta Bounding Hot Case

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Abstract

This report documents the preliminary passive thermal design and radiator sizing trade-off analysis of a high-power 6U technology-demonstration CubeSat in a circular 600 km dawn-dusk Sun-Synchronous Orbit (SSO). In dawn-dusk SSO configurations, high solar beta angles during solstices drive near-continuous solar exposure, presenting extreme over-temperature challenges for internal avionics and battery packs. We present **Passive Redesign v2**, which eliminates the need for active cooling or deployable mechanisms. The redesign uses passive thermal zoning, surface coating selection (AZ-93 silicate paint and silvered FEP tape), and orbit-attitude-dependent radiator placement on the anti-Sun cross-track face. A dual-axis system-level radiator sizing sweep reveals that a 0.02 m² radiator offers the optimal balance between hot-case thermal safety margin (avionics peak at 39.2°C) and cold-case heater power penalties (5.92 Wh/orbit).

1 Introduction

Modern small satellites are increasingly packed with high-power payloads and avionics, leading to significant internal power dissipation. A standard 6U CubeSat (10 cm × 20 cm × 30 cm) technology-demonstrator requires robust thermal architecture to operate under bounding orbital conditions.

This study focuses on a 6U spacecraft in a circular 600 km dawn-dusk SSO with an inclination of $\approx 97.8^\circ$. Under solstice conditions, the orbit plane tilts relative to the sun vector, yielding a high beta angle ($\beta = 74^\circ$) that exceeds the critical shadow angle ($\approx 66.07^\circ$), causing the satellite to remain in continuous sunlight. The SSO geometry and sunlit schedule are shown in Figure 1.

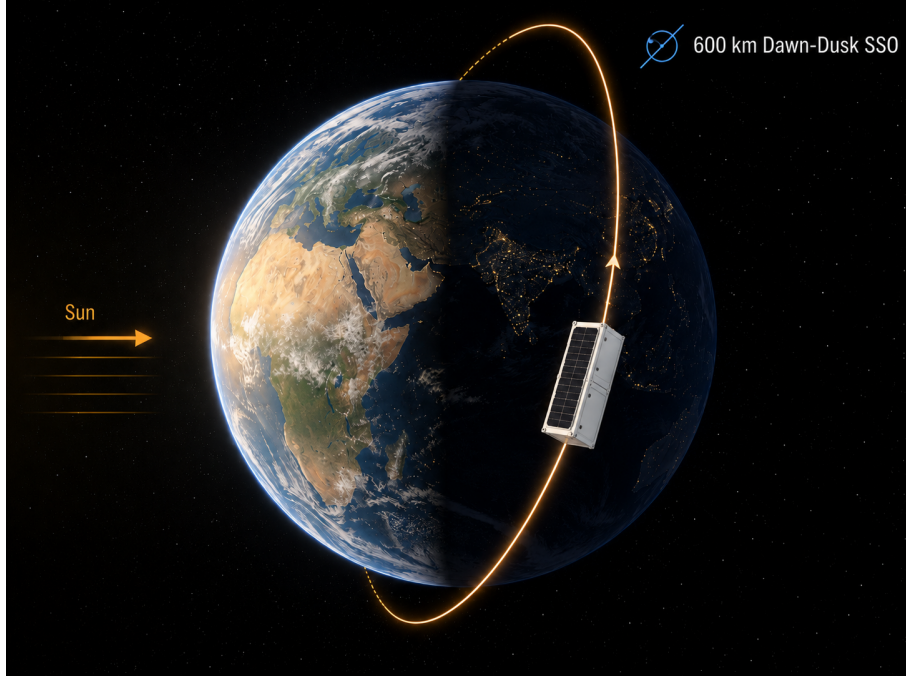


Figure 1: Circular 600 km Dawn-Dusk SSO Orbit Geometry, illustrating the permanent solar illumination vector on the cross-track panel.

2 Thermal Limits and Component Specifications

Thermal node mass, capacitance, and operational limits are traceable to engineering standards. The component details are listed in Table 1.

Table 1: Component Mass, Capacitance, and Temperature Limits

Node ID	Mass [kg]	c_p [J/kg·K]	Min Limit [°C]	Max Limit [°C]	Internal Diss. [W]
Battery	1.5	800.0	-10.0	45.0	0.5
Avionics	1.0	900.0	-20.0	60.0	10.0 – 15.0
Payload	2.0	900.0	-10.0	50.0	0.0 – 15.0
Structure	3.0	900.0	-40.0	80.0	0.0
Radiator	0.5	900.0	-100.0	100.0	0.0
External Shell	1.5	900.0	-100.0	100.0	0.0

The 6-node lumped-parameter thermal network architecture, conductive couplings G_{ij} , and external load interfaces are shown schematically in Figure 2.

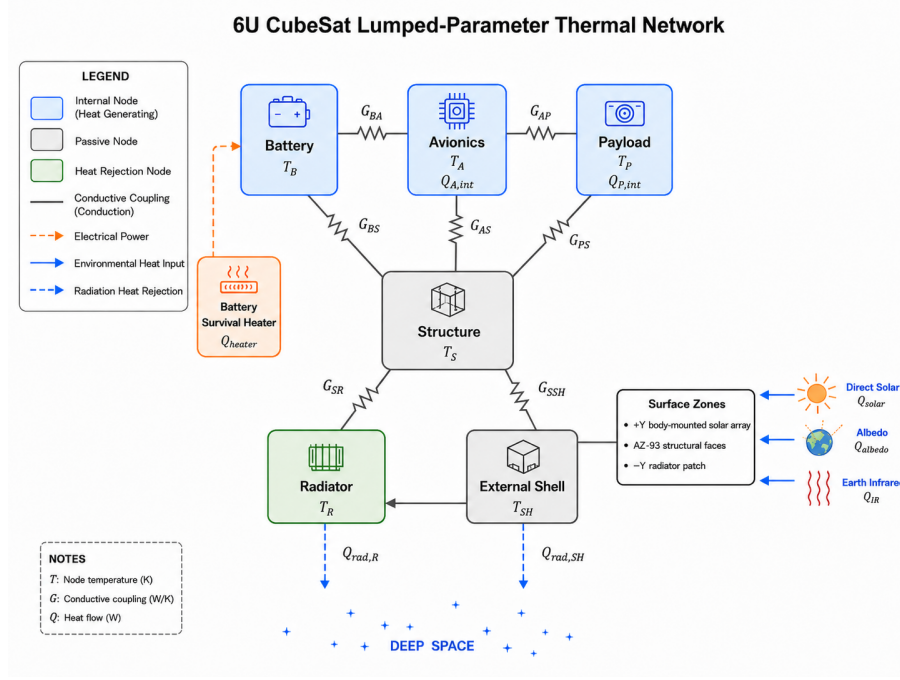


Figure 2: 6-Node Lumped Parameter Thermal Network diagram, showing inter-node conductive paths and environmental radiation links.

3 Material Library and Optical Properties

All thermo-optical coatings used in this design are canonical candidates sourced directly from the NASA SmallSat State of the Art (SoA) Thermal Control webpage, specifically Tables 7-3 and 7-4. The material properties catalog is summarized in Table 2.

Table 2: Material Library and BOL Optical Properties

Material ID	Intended Role	α_s (BOL)	ϵ_{IR}	NASA Source	Concise Description
AZ-93 Silicate	External shell	0.16	0.90	Table 7-3	Inorganic white thermal paint
0.005" FEP/Ag/Inconel	Primary radiator	0.07	0.79	Table 7-4	Second-surface silvered FEP tape
VDA/200HN/PSA	Reference insulation	0.08	0.03	Table 7-4	Metallized tape, very low emissivity
Z306 Polyurethane	Comparison paint	0.92	0.89	Table 7-3	Matte black polyurethane paint
solar_cell_assembly	Solar panels	0.73	0.82	Section 7.1	Body-mounted solar cells

4 Baseline Failure Analysis

The initial baseline configuration mapped high-absorptance black paint ($\alpha = 0.92, \epsilon = 0.89$) to several outer shell faces representing solar panel substrates. Under the bounding hot case ($\beta = 74^\circ$), the large surface absorption caused extreme solar heat loading. This resulted in catastrophic thermal limits breach across multiple nodes (Battery peaking at 80.6°C , Avionics peaking at 81.6°C).

Additionally, using a low-emissivity material (like metallized VDA tape, $\epsilon = 0.03$) as a radiator was shown to cause thermal traps, raising avionics to 104.1°C due to the inability to reject heat to deep space.

5 Passive Redesign v2 Architecture

To resolve these violations passively, we implemented three key solutions in Redesign v2:

- **Thermal Surface Zoning:** We mapped AZ-93 white paint ($\alpha = 0.16, \epsilon = 0.90$) to five non-power shell faces to minimize solar absorption while maximizing deep-space emittance.
- **Radiator Face Orientation Physics:** In dawn-dusk SSO, the cross-track $+Y$ face is permanently Sun-facing, while the opposite $-Y$ face is permanently shaded (anti-Sun). The dedicated radiator (0.02 m^2 silver FEP tape, $\alpha = 0.07, \epsilon = 0.79$) was relocated to the anti-Sun $-Y$ face to achieve the lowest integrated environmental heat load.
- **Solar Cell Electrical Offset:** Since 18% of the absorbed solar energy on the solar arrays is converted to electricity and exported out of the thermal network, we applied an electrical conversion offset:

$$\alpha_{\text{eff}} = \alpha_s - \eta_{\text{elec}} = 0.73 - 0.18 = 0.55 \quad (1)$$

Figure 3 illustrates the surface coating allocations and zoning layout for the redesigned spacecraft.

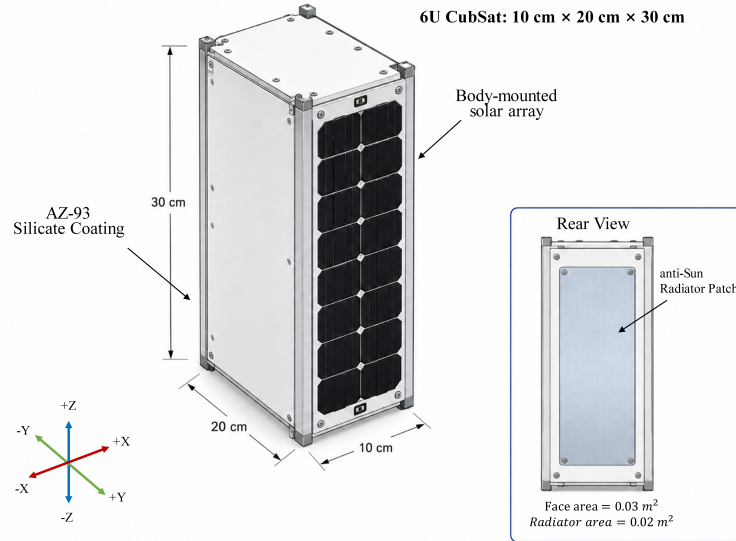


Figure 3: Spacecraft Passive Thermal Zoning layout (Redesign v2), showing body-mounted solar cell arrays on the Sun-facing $+Y$ face, AZ-93 white paint on the general shell, and the silver FEP radiator on the anti-Sun $-Y$ face.

6 Simulation Results

Applying the Redesign v2 architecture, all scenarios successfully converged to PASS conditions. Table 3 summarizes the steady-state node temperatures.

Table 3: Steady-State Temperature Ranges under Redesign v2

Node ID	Nominal ($\beta = 65^\circ$) [$^\circ\text{C}$]	Bounding Hot ($\beta = 74^\circ$) [$^\circ\text{C}$]	Cold Case ($\beta = 60^\circ$) [$^\circ\text{C}$]
Battery	14.8 to 15.1	35.0 to 35.4	5.0 to 10.0
Avionics	16.4 to 17.1	38.1 to 39.2	-0.8 to 0.8
Payload	14.1 to 17.7	34.0 to 40.3	-1.7 to -0.5
Structure	13.2 to 14.2	33.5 to 34.8	-2.2 to -0.1
Heater Duty	0.0%	0.0%	36.7%
Heater Energy	0.0 Wh	0.0 Wh	5.92 Wh/orbit

7 Radiator Sizing System-Level Trade-Off

A sweep of radiator area A_{rad} from 0.02 to 0.06 m² was performed across all three scenarios to evaluate the trade-off between hot-case avionics peak temperature and cold-case heater power draw (Table 4).

Table 4: Radiator Area Sweep Trade-Off Results

Area [m ²]	Hot Peak Avionics [$^\circ\text{C}$]	Cold Min Battery [$^\circ\text{C}$]	Cold Heater Duty [%]	Cold Heater Energy [Wh]
0.02	39.17	5.00	36.7%	5.92
0.03	34.87	5.00	41.5%	6.69
0.04	31.15	5.00	49.0%	7.90
0.05	27.80	5.00	63.6%	10.26
0.06	24.81	4.99	60.9%	9.81

Increasing the radiator size significantly drops the hot-case peak temperature, but imposes a severe 73% heater energy penalty in the cold case. Due to strict power limitations on small satellites, the **0.02 m² radiator size is selected as the optimum thermal design**.

8 Visualizations and Design Analysis

The transient temperature response and parameter sizing trade-offs are visualized below. Figure 4 illustrates the nodal temperature histories under the continuous sunlight bounding hot case, showing stable convergence. Figure 5 depicts the dual-axis trade-off between peak hot avionics temperatures and cold battery margins across various radiator areas.

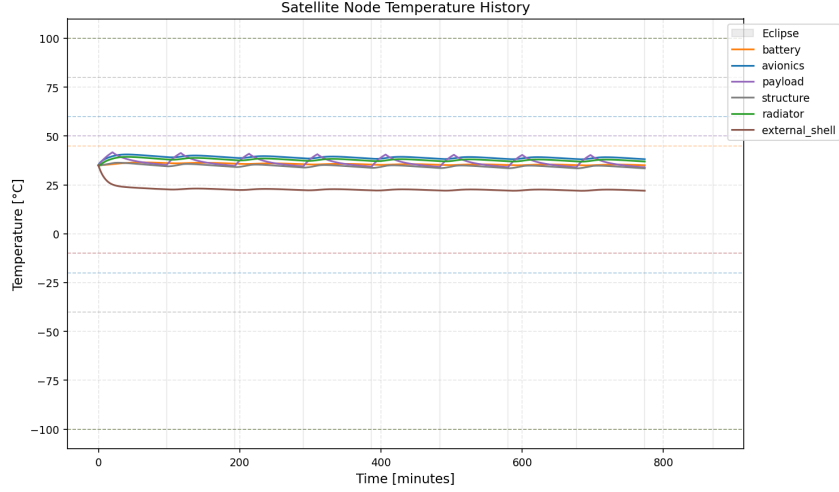


Figure 4: Nodal Temperature Histories Under Bounding Hot Case ($\beta = 74^\circ$, Continuous Sun-light Dawn-Dusk SSO Solstice).

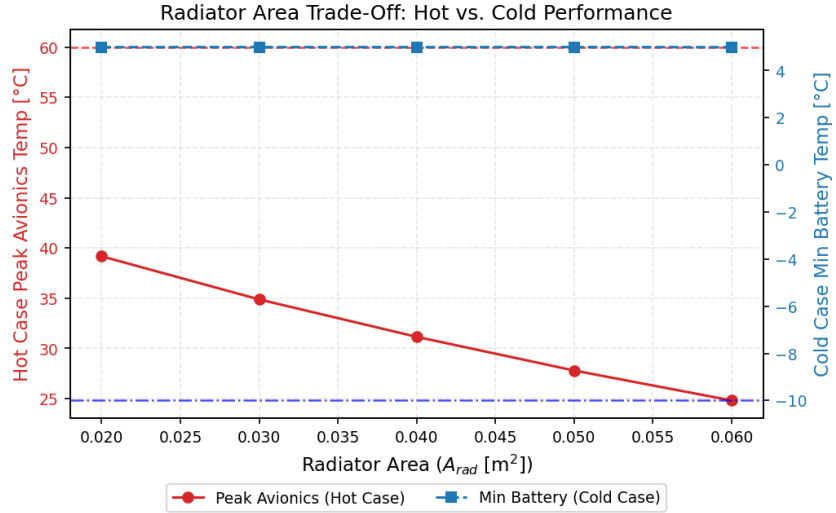


Figure 5: Dual-Axis Radiator Sizing Trade-Off (Hot Case Peak Avionics Temp vs. Cold Case Min Battery Temp).

9 Conclusion

This report demonstrates that a rigorous passive thermal redesign utilizing surface zoning, anti-Sun radiator placement, and electrical power offset calculations successfully resolves the hot-case over-temperature limits of a 6U CubeSat in 600 km dawn-dusk SSO. Sizing trades highlight the importance of balancing radiator design to minimize cold-case heater power penalties.